A defined AF forecasting prototype

Clinical history plus dynamic ECG; clinician review; research only.

Forecast target

Previous 600 seconds; new AF onset 120-600 seconds ahead; update every 30 seconds. AF is different from heart attack or stroke.

Challenge fusion

The declared challenge finalist is logistic_combined: paired IRIDIA age/sex and causal ECG. An ECG-only model displayed beside demographic fields would not establish fusion.

Scope

A persistent virtual patient shows changing baseline, signal quality, risk and warning history. Historical replay is labeled; model weights stay frozen. No automated treatment decisions.

Audited real data and limitations

IRIDIA-AF primary cohort; LTAF reserved external transfer.

Paired primary cohort

IRIDIA: 167 recordings / 152 people, 200 Hz two-lead ECG and baseline age/sex. Development: 132 recordings / 121 groups, 5,243.98 monitored hours and 241 eligible AF onsets.

External transfer

LTAF: 84 two-lead recordings, 128 Hz, reviewed .atr rhythm references. Missing clinical fields remain missing. Recording IDs do not establish unique people or cross-dataset subject separation.

Timing safeguards

No invented diagnoses or unrelated synthetic EHR joins. Prior-only features; minimum AF duration 30 s; washout 600 s. IRIDIA daily calibration intervals excluded; upstream measurement limitations remain.

Measured development comparisons

Same grouped development cohort; these are not held-out test results.

Fused challenge finalist

Combined logistic: 6/241 events (2.49%), FAR 0.1706/eligible h, 761 starts, 40.78% warning occupancy. It fails the 50% engineering sensitivity gate; fusion benefit is not established.

Scientific reference preserved

ECG current recipe: 7/241 events (2.90%), FAR 0.1186/eligible h, 624 starts, 21.84% occupancy. The new challenge constraint was declared after development and before held-out access.

Completed CUDA challenger

RR TCN16: 4/241 events, FAR 0.0300/eligible h, 154 starts, 0.088% occupancy; 9,092.06 s full candidate cost. Waveform completed only 2/3 folds and is excluded; RR32 was not reached.

Persistent patient twin and replay

State adapts to prior observations; the chosen classifier omits baseline-deviation inputs.

Two input streams

Paired clinical values and causal ECG feed the fused predictor. Verify the actual frozen schema, coefficients and controlled input sensitivity; displaying fields alone is insufficient.

Changing state

Trailing ECG/RR features, warm-up, quality, baseline and forecast evolve along historical time. SQLite commits cursor, baseline, warning and review state atomically with restart identity guards.

Working demonstration

Show artifact-bound real-prefix research replay logs. Label the manufactured-data dashboard as the conceptual doctor UI; it does not serve this failed-gate artifact. Held-out warnings and misses come from offline reports.

Frozen primary patient test

Saved evaluation only; no test-set threshold or model retuning.

Clinical + ECG finalist

1/43 events; sensitivity 2.3% (95% CI 0.0-7.9%); FAR 0.0850/eligible h; 113 starts; occupancy 1.7%; coverage 92.9%; group Brier 0.006657.

Preserved ECG scientific reference

0/43 events; sensitivity 0.0% (95% CI 0.0-0.0%); FAR 0.0000/eligible h; 0 starts; occupancy 0.0%; coverage 92.9%; group Brier 0.006661.

Interpretation and provenance

300 patient-group bootstrap draws; the reference [0,0] interval is degenerate, not precise safety evidence. Frozen model/split/prediction hashes bind these results. Full denominators, misses and lead time are in the report.

Frozen external LTAF transfer

External inputs cannot change preprocessing, calibration, weights or policy.

Clinical + ECG finalist

2/29 events; sensitivity 6.9% (95% CI 0.0-16.7%); FAR 0.0083/eligible h; 398 starts; occupancy 6.2%; coverage 89.4%; group Brier 0.012055.

Preserved ECG scientific reference

0/29 events; sensitivity 0.0% (95% CI 0.0-0.0%); FAR 0.0000/eligible h; 0 starts; occupancy 0.0%; coverage 89.4%; group Brier 0.012054.

Transfer limits

392/398 fusion starts are unassessable: 299 known-excluded, 93 unknown. At 60/300 s minimum AF duration, fusion detects 0/19 and 0/8. Clinical fields are missing; record-group intervals do not prove independent people or Indian generalization.

Warnings, uncertainty and failure

Small start-rate budgets do not guarantee a low warning-time burden.

Policy

Require usable history/quality and persistent elevated calibrated risk. Hysteresis and cooldown suppress duplicate starts. Count every eligible onset, including misses during abstention.

Honest measurement

Report false starts per eligible hour, total starts, occupied time, coverage, excluded/unknown warnings, calibration and lead time together. Low Brier in rare events does not establish useful forecasting.

Acceptance

Both declared finalists fail development acceptance. Default serving refuses failed-gate artifacts. Research evaluation can still quantify limitations; neither clinician readiness nor improved outcomes is claimed.

Compute cost and debugging evidence

Allocated GPU hardware and useful compute are separate measurements.

GPU use verified

Saved fit receipts and placement code confirm CUDA use on one device despite two T4s being available. The neural screen consumed 20,884.37 s; its candidate/fold costs are nested and must not be added twice.

Why the run was slow

The 10,000-window cap applies to fitting. Normalization, source IO, calibration, policy search and 604,231 available outer decisions also cost time; much preprocessing runs on CPU.

Diagnostics

Current source emits timestamped stage start/done/failure, bounded progress counters, actual model device and NVIDIA utilization samples. These changes were absent from the immutable completed Kaggle v1 run; no T4 speedup benchmark is claimed.

Software checks and reproducibility

Engineering evidence is separate from clinical forecasting evidence.

178 passing tests

Recorded diagnostics suite: 178 tests passed. Coverage includes leakage/group guards, target boundaries, alert accounting, causal state, restart and progress failures. Synthetic fixtures validate software, not clinical sensitivity.

Reproduce

Locked Python/npm dependencies, clean base-only wheel verification, optional neural dependencies, saved source/data/split/model hashes and recorded runtime versions. CUDA versus MPS is confounded by runtime differences.

Practical limits

Local API/React dashboard and SQLite; bind to localhost. No hospital authentication, PHI governance or prospective wearable validation. Raw data and credentials stay outside the public repository.

Sources, delivery and next validation

One honest challenge solution; real identity and recording supplied at final delivery.

Data attribution

Gilon et al. (2023), IRIDIA-AF v1.0.1: zenodo.org/records/8405941, CC BY 4.0. Petrutiu et al. (2007), LTAF v1.0.0: physionet.org/content/ltafdb/1.0.0/, ODC Attribution 1.0. Code: MIT; full citations in DATA_LICENSE.md.

Required delivery

Public GitHub, actual team/college, architecture/presentation PDFs and a real 20:00-minute unlisted YouTube demo linked in README. Identity, recording and submission remain pending. Register 18 Oct; submit 20 Oct 2026, both 19:00 IST.

Clinical next step

Ethically approved Indian clinical partner data, device/patient-group calibration and prospective silent evaluation before care use. Official rules: unstop.com/hackathons/crp-digital-twin-challenge-2026-happiest-health-1757873.